# Topic 1, 2 and 3. Classes and Objects. Static. Final. Enum. BigDecimal. LocalDate. Abstract Classes. Interfaces. Exceptions. Collections. Serialization

## Shop Project

In the shop, various types of **goods** need to be delivered and sold. Each of them is defined by an **ID, name and category** (food or non-food).

### Selling Price Calculation

- Food and non-food goods have different **markup percentages**, which are determined by the shop.
- If a product is **close to its expiry date** (i.e., fewer than a given number of days remain), its selling price is reduced by a **specified discount percentage**.
- The **number of days** before expiration that triggers a discount and the **discount percentage** vary for each shop.
- Expired goods **cannot be sold**.

### Cashiers and Cash Desks

- Cashiers work in the shop and are identified by their **name and ID**.
- Each cashier is assigned to a **cash desk**. There are customers that wait at each cash desk to buy the products in their basket.
- If the customer has **enough money**, the cashier issues a **receipt**.

### Stock Management

- Goods can be sold **only if there is sufficient stock** in the shop.

### Receipt Details

Each **receipt** includes:

- **Receipt ID**
- **Cashier** who created it
- **Date** of purchase
- Quantity and price of each goods
- **Total amount** the customer must pay
- The receipt has to be **serialized** and **deserialized**

- The receipt has to be stored in a **.txt file** that includes the id of the receipt in the name of the file

## Financial Calculations for Each Shop

For every shop, the following financial metrics must be calculated:

1. **Total costs** – including **cashier salaries** and **delivery costs of goods**
2. **Total income** – revenue from **sold goods**
3. **Profit** – calculated as **income minus costs**

Implement Exception Handling and use appropriate collections!